

University of Central Florida

Department of Computer Science

COP 3402: Systems Software

Spring 2024

Final Exam (April 30th)

(This exam has 9 questions)

Read This Carefully

All answers must demonstrate an understanding of problem and solution. Please show your work if you would like to receive partial credit. Explanations that are incorrect will affect your grade even if correct explanations are also present.

State your assumptions as needed. They should not be unreasonable or lead to a trivial solution. When in doubt, ask the instructor.

Write your name on first page.

The exam begins at 10:00 a.m. and ends at 12:00 p.m. (Noon)

Then, you will have from 12:01 p.m. to 12:15 p.m. to submit

After 12:15 p.m. exams will be accepted by email with a penalty of (-20)

After 12:20 p.m. exams will be accepted with a penalty of (-30)

After 12:25 p.m. your grade is zero

(No excuses accepted)

You will have from 10:00 a.m. to 10:10 a.m. to ask questions (via Zoom).

Giving or receiving aid in an exam constitutes a violation of the UCF Golden Rule and may result in penalties as described therein

Exam is open-book, open Lectures, and you can use any device you need.

Name: _____

Question	1	2	3	4	5	6	7	8	9
Points									

1.- [12 points] When generating code for instructions LOD and STO (in your compiler) a lexicographical level must be generated to indicate whether the top of the stack value is loaded or stored locally (current activation record) or it has to be stored to or loaded from another activation record.

a) How is the lexicographical level calculated in instructions LOD, STO, and INC?

b) How does your compiler keep track of the current lexicographical level?

2.- [12 points] On Operating Systems

a) What is an Operating System?

b) What are the three main characteristics of an Operating System?

3.- [12 points] On interrupts:

a) When a running process initiates an I/O operation, an interrupt occurs. What is the name of the interrupt?

b) Assume that process P1 is in the waiting for I/O state, P3 is in the ready state, and process P2 is running. In this scenario, an Input/Output interrupt, a timer interrupt, and a system call (SIO) occur simultaneously. The OS must handle all interrupts. Describe in one short sentence each step carried out by the OS.

4. [12 points] Process Synchronization.

A running process executes a P(S) operation. Does it block? (Justify your answer)

5.- [12 points] The run time environment is composed of five segments: CODE(TEXT), DATA, BSS, HEAP, and STACK. You have to tell in what segment you will place each one of the following examples:

- a) `int seven = 7;` →
- b) `seven = seven + 7;` →
- c) `seven = 7;` →
- d) `int seven;` →
- e) `malloc(seven);` →

6. [12 points] Draw a diagram showing the ELF (object code), Process Control Block, Thread Control Blocks, and the run-time environment for a program with two threads, Indicating: how these data structures are related. Use arrows to show all relationships. Name all fields of the TCB and at least 2 of the PCB.

Hint: Assume the object code file has three sections plus header.

7.- [12 points] Draw a diagram showing the states a process can be in and indicate what is the event (interrupt) that moves the process from one state to another. Use the states explained in class. If you use a new state, which was not explained in class, you must explain how the new interrupts work.

8.- [12 points] Assemblers and code generation.

Assume you are given the following program in PL/0 to be compile in your HW3:

```
var else, x, call;  
x := call + 5 * else. (Parse the expression left to right and then, store the resulting value in x)
```

Translate it into assembly language. Use the ISA of your virtual machine.

9.- [4 point] What is an EOC interrupt?